

Counterfactual Inflation Analysis: What If Ukraine Had Been Part of the Euro Area?

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Abstract

This report studies Ukraine’s monetary regime and constructs a counterfactual inflation path under the hypothetical scenario in which Ukraine had been part of the Euro Area. Part A documents Ukraine’s exchange-rate regime from 2000 to 2025, identifying de facto dollar pegs, crisis devaluations, capital-control episodes, the transition to inflation targeting, and the wartime fixed exchange-rate regime. This chronology is central to the identification strategy because Euro Area membership does not represent a homogeneous treatment over time: its marginal effect is limited during peg periods but stronger during devaluation episodes and periods of greater monetary-policy autonomy. Part B uses this time-varying treatment logic to estimate counterfactual Ukrainian inflation with an ARDL/local-projection specification, a Blanchard–Quah SVAR robustness check, and a synthetic-control benchmark.

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1 Ukraine’s Monetary Regime: Preliminary Analysis

Before constructing a counterfactual inflation path, it is necessary to identify what Ukraine actually did with its monetary sovereignty. Treating Ukraine as a freely floating economy over the entire 2000–2025 sample would be misleading. For most of the pre-2014 period, the hryvnia was de facto anchored to the US dollar, first around 5 UAH/USD and then around 8 UAH/USD. The main regime breaks are the 2008–2009 Global Financial Crisis, the collapse of the peg in 2014–2015, the transition to inflation targeting in 2015–2016, and the wartime fixed exchange-rate regime introduced after the full-scale invasion in 2022. This distinction between de jure and de facto monetary sovereignty is precisely the issue emphasised by the fear-of-floating literature ([Calvo and Reinhart, 2002](#)). It is central for the counterfactual exercise: euro-area membership should not be treated as a homogeneous shock over time, because during peg periods Ukraine had already given up much of its effective monetary autonomy.

1.1 Exchange-rate regime chronology

Table 1 summarises Ukraine’s exchange-rate regime chronology over 2000–2025. The classification combines the observed UAH/USD path, NBU exchange-rate and policy-rate data, IMF AREAER-style terminology, IMF Ukraine country reports, and official NBU monetary-policy communications ([National Bank of Ukraine, 2025a,b](#); [International Monetary Fund, 2014, 2015, 2024, 2025](#)).

Table 1: Ukraine's exchange-rate regime chronology, 2000–2025

Period	Approx. UAH/USD rate	De facto regime classification	Key events and interpretation
2000–2006	around 5.2 to 5.0	De facto peg / conventional peg	The hryvnia remained tightly anchored to the US dollar. IMF AREAER-style classifications identify Ukraine as a conventional peg in the mid-2000s, and exchange-rate data show a long period of stability around 5 UAH/USD. Monetary-policy autonomy was therefore very limited.
2007–Sept. 2008	around 5.0	Quasi-peg / managed float de jure	Although the IMF classification moved toward managed floating, the exchange rate remained close to 5 UAH/USD until the Global Financial Crisis. Formal flexibility existed, but actual monetary sovereignty remained constrained by the exchange-rate objective.
Oct. 2008–2009	from about 5.8 to about 8.0	Other managed arrangement / crisis devaluation	The Global Financial Crisis triggered a sharp depreciation, reserve pressure, banking stress, and balance-of-payments tensions. Crisis-related restrictions were introduced to reduce capital outflows and preserve financial stability (International Monetary Fund, 2014).
2010–Feb. 2014	around 8.0; 7.993 from July 2012	Stabilized arrangement / renewed de facto USD peg	After the 2008–2009 depreciation, the official exchange rate was re-stabilised around 8 UAH/USD. IMF reports describe this period as a de facto peg or stabilised arrangement. Monetary policy was again subordinated to exchange-rate stability.
Mar. 2014–2015	from about 8 to above 20; stabilisation around 24–25 later	Peg collapse / crisis float under restrictions	In February 2014, the NBU abandoned the peg as reserves became insufficient. The Crimea annexation, Donbas conflict, reserve losses, and the confidence shock generated a very large depreciation. The adjustment was accompanied by administrative restrictions, capital controls, and a fragmented FX market (International Monetary Fund, 2015).
2016–2021	mostly mid-20s, with two-way fluctuations	Managed float / inflation-targeting framework	Ukraine entered an inflation-targeting regime with a more flexible exchange rate. The transition was prepared in 2015 and formalised in 2016. The NBU's key policy rate became a genuine monetary-policy instrument, although foreign-exchange intervention did not disappear (National Bank of Ukraine, 2016, 2026).
24 Feb. 2022–20 July 2022	fixed around 29.25	Wartime hard peg / stabilised arrangement	At the start of the full-scale invasion, the NBU suspended normal FX-market functioning, fixed the official exchange rate, limited cash withdrawals, and imposed restrictions on many cross-border payments. Monetary sovereignty was formally national, but practically dominated by emergency wartime stabilisation (National Bank of Ukraine, 2026).
21 July 2022–2 Oct. 2023	fixed at 36.5686	Stabilized arrangement	On 21 July 2022, the NBU devalued the official exchange rate by 25% and fixed it at 36.5686 UAH/USD. The regime remained a wartime peg with strict FX restrictions and capital controls (National Bank of Ukraine, 2022).
Oct. 2023–2025	above the wartime peg, with gradual depreciation	Managed flexibility; IMF de facto floating from 3 Oct. 2023	On 3 October 2023, the NBU moved to managed exchange-rate flexibility. IMF reports later reclassified Ukraine's de facto exchange-rate arrangement from stabilised to floating, effective from 3 October 2023. In practice, the regime remained heavily managed, with interventions and wartime FX restrictions still relevant (National Bank of Ukraine, 2023 ; International Monetary Fund, 2024, 2025).

Figure 1 plots the monthly average UAH/USD exchange rate. The flat segments around 5 and 8 UAH/USD identify the main pre-2014 dollar-anchor periods. The largest regime break occurs in 2014–2015, while 2022–2023 combines an emergency wartime peg with an official devaluation.

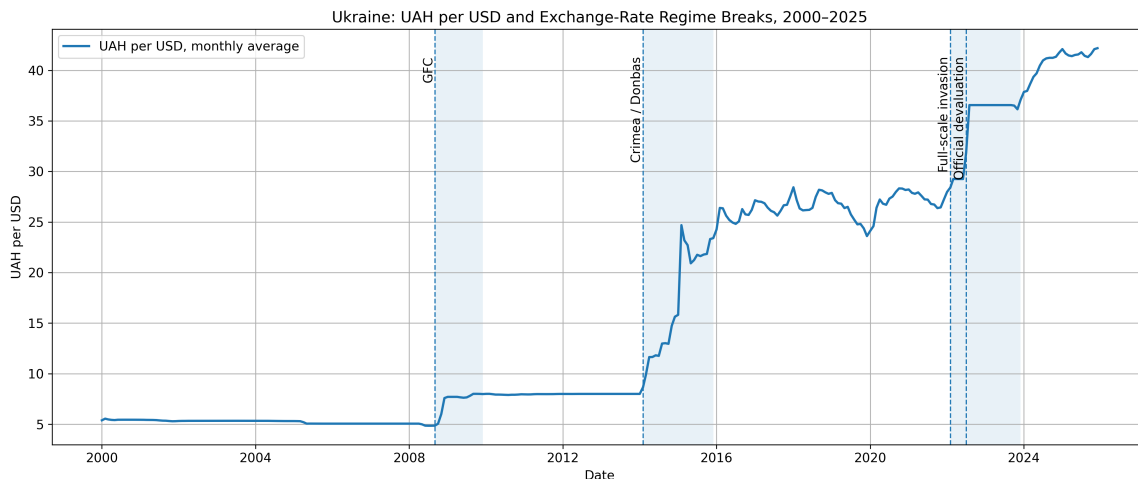


Figure 1: Ukraine: UAH per USD and exchange-rate regime breaks, 2000–2025

1.2 Major devaluations and capital controls

Three devaluation episodes are central for the counterfactual analysis. The first is the 2008–2009 Global Financial Crisis, when the hryvnia moved from the quasi-peg around 5 UAH/USD to around 8 UAH/USD. The second is the 2014–2015 crisis, when the NBU abandoned the peg and the exchange rate moved from around 8 UAH/USD to above 20 UAH/USD amid the Crimea annexation, the Donbas conflict, reserve losses, and a collapse in confidence. The third is the 2022 wartime episode, when the NBU first fixed the exchange rate at 29.25 UAH/USD, then devalued it by 25% to 36.5686 UAH/USD on 21 July 2022 ([National Bank of Ukraine, 2022](#)).

Table 2: Major hryvnia devaluation episodes

Episode		Start	End	Start rate	End rate	Window dep.	Trigger
2008–2009	Global Financial Crisis	2008-06	2009-12	4.85	7.98	64.43%	Global financial crisis, capital outflows, banking-sector stress, and balance-of-payments pressure.
2014–2015	Crimea / Donbas crisis	2013-12	2015-12	7.99	23.41	192.86%	Crimea annexation, Donbas conflict, reserve losses, collapse of confidence, and macro-financial stress.
2022–2023	Full-scale invasion	2021-12	2023-12	27.21	37.09	36.31%	Full-scale invasion, wartime fixed exchange rate, official devaluation, and strict FX restrictions.

Capital controls were not equally important throughout the sample. In 2008–2009, they

were crisis-management tools. In 2014–2015, they became a central stabilisation device, as NBU and IMF documents show a tightening of administrative restrictions during the exchange-rate collapse ([International Monetary Fund, 2015](#)). These measures were gradually relaxed after stabilisation, especially after the more liberal foreign-exchange framework introduced in 2019. In 2022–2025, by contrast, FX restrictions and capital controls became constitutive features of the wartime monetary regime: the NBU fixed the exchange rate, restricted cash withdrawals and cross-border payments, and only gradually loosened controls after the 2023 move toward managed flexibility ([National Bank of Ukraine, 2023](#); [International Monetary Fund, 2024](#)).

Figure 2 confirms that depreciation pressure is concentrated in specific crisis windows rather than evenly distributed over time.



Figure 2: Ukraine: monthly hryvnia depreciation spikes, 2000–2025

1.3 Policy-rate behaviour and inflation targeting

The transition to inflation targeting has two relevant dates. In 2015, the NBU prepared the transition by identifying inflation targeting as the appropriate future framework. The official disinflation path was $12 \pm 3\%$ by end-2016, $8 \pm 2\%$ by end-2017, $6 \pm 2\%$ by end-2018, and $5 \pm 1\%$ from 2019 onward ([National Bank of Ukraine, 2016](#)). In December 2016, the NBU Council formally approved the inflation-targeting regime, although the NBU describes the practical shift as having started already in early 2016 ([National Bank of Ukraine, 2026](#)).

This was not only an institutional rebranding. Before 2015, the key policy rate had a limited operational role because the exchange-rate objective dominated monetary policy. After the 2015–2016 operational reform and the move away from the hard peg, the policy rate became a genuine monetary-policy instrument, influencing money-market rates, the exchange rate, and inflation expectations. Figure 3 illustrates this shift: policy-rate adjustments became more meaningful after the 2014–2015 crisis and during the inflation-targeting

period.

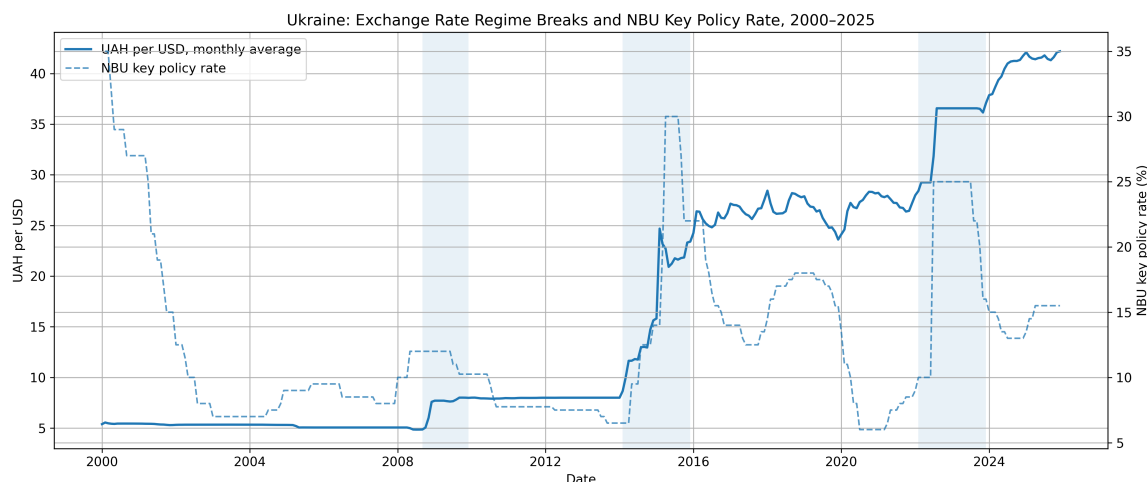


Figure 3: Ukraine: exchange-rate regime breaks and NBU key policy rate, 2000–2025

The war interrupted full-fledged inflation targeting. In 2022, the NBU temporarily prioritised nominal and financial stabilisation through a fixed exchange rate, FX restrictions, and administrative measures. From October 2023 onward, Ukraine moved toward managed flexibility, but this should not be interpreted as a full return to normal inflation targeting. The more accurate description is a gradual return toward inflation targeting under wartime constraints ([National Bank of Ukraine, 2026, 2023](#)).

1.4 Implication for the counterfactual treatment

The key implication is that Ukraine’s monetary sovereignty was intermittent. It was very limited during the long de facto dollar-peg periods, especially 2000–2008 outside the crisis and 2010–February 2014, because the exchange-rate objective dominated monetary policy. It increased during regime breaks and more flexible periods, especially in 2008–2009, 2014–2015, 2016–2021, and, more cautiously, after October 2023. However, these episodes of greater autonomy often coincided with crisis, capital controls, heavy intervention, or wartime restrictions. Therefore, hypothetical euro-area membership should not be modelled as a constant treatment. Its marginal effect would have been small when Ukraine was already following a quasi-dollar peg, but large when the exchange rate acted as an adjustment buffer or when the NBU was genuinely using the policy rate to target inflation. This is the de facto sovereignty logic implied by [Calvo and Reinhart \(2002\)](#): formal monetary sovereignty matters less than the actual constraints imposed by the exchange-rate regime.

Sources. Exchange-rate and policy-rate data are taken from the National Bank of Ukraine ([National Bank of Ukraine, 2025a,b](#)). Regime classifications are interpreted using IMF AREAER terminology and IMF Ukraine country/program reports ([International Monetary Fund, 2014, 2015, 2024, 2025](#)). The inflation-targeting dates and target path are based

on NBU monetary-policy communications ([National Bank of Ukraine, 2016, 2026](#)). The distinction between formal and effective monetary sovereignty follows the fear-of-floating framework of [Calvo and Reinhart \(2002\)](#).

2 Counterfactual Inflation Analysis

2.1 Objective

The objective of Part B is to construct a counterfactual year-on-year inflation series for Ukraine under the hypothesis that Ukraine had been a member of the Euro Area. The counterfactual cannot be a simple cross-sectional average of Euro Area inflation. It must rely on an explicit identification strategy and remain consistent with the monetary-regime chronology established in Part A.

The analysis uses three complementary methods. The baseline specification is a monthly ARDL/local-projection-style model estimated over the full sample from 2001 to 2025. This specification is preferred because it covers the three main crisis episodes required by the exam: the 2008–2009 Global Financial Crisis, the 2014–2015 Crimea/Donbas crisis, and the 2022 full-scale invasion. As a structural robustness check, I estimate a Blanchard–Quah SVAR in the spirit of [Blanchard and Quah \(1989\)](#) and [Bayoumi and Eichengreen \(1993\)](#). Finally, I use a synthetic-control benchmark following [Abadie et al. \(2010\)](#).

2.2 Data

Ukraine’s actual inflation is computed from the raw monthly CPI data. The original Ukrainian CPI observations are monthly index changes, with the previous month equal to 100. I convert these observations into monthly price-level factors and compute year-on-year inflation as the rolling twelve-month cumulative product:

$$\pi_t^{UA} = \left(\prod_{j=0}^{11} \frac{CPI_{t-j}^{UA}}{100} - 1 \right) \times 100. \quad (1)$$

The Euro Area inflation block is constructed from a panel of HICP year-on-year inflation rates for eleven Euro Area countries: Germany, France, Italy, Spain, the Netherlands, Belgium, Austria, Portugal, Ireland, Finland, and Greece. Instead of using a mechanical cross-sectional mean, I extract a common Euro Area inflation factor using principal component analysis, following the logic of [Ciccarelli and Mojon \(2010\)](#). The first principal component explains approximately 80% of the cross-country variance in the Euro Area inflation panel.

For the main ARDL/local-projection model, I also use the UAH/EUR exchange rate from the National Bank of Ukraine and compute year-on-year log depreciation:

$$\Delta e_t^{yoy} = 100 [\log(E_t) - \log(E_{t-12})], \quad (2)$$

where E_t is the number of hryvnias per euro. A positive value therefore represents a depreciation of the hryvnia against the euro.

For the BQ-SVAR robustness exercise, I use Ukraine industrial-production year-on-year growth as the Ukrainian activity variable and Euro Area industrial-production year-on-year growth as the Euro Area activity variable. The use of activity and inflation follows the structural-shock decomposition logic of [Blanchard and Quah \(1989\)](#) and [Bayoumi and Eichengreen \(1993\)](#).

All external series used in the analysis are either provided with the replication files or documented in the code. The external variables are used for identification rather than description only: the exchange-rate series identifies the depreciation channel in the ARDL/LP model, while industrial-production growth provides the activity variable required for the Blanchard–Quah SVAR. This satisfies the requirement that the counterfactual be based on macroeconomic variables beyond inflation alone.

2.3 Treatment intensity from Part A

The treatment intensity is directly derived from the Part A monetary-regime chronology. The central idea is that Euro Area membership would have mattered little during periods when Ukraine was already de facto constrained by a dollar peg, but would have mattered strongly during episodes of genuine monetary autonomy, floating exchange rates, and large devaluations.

Let $w_t \in [0, 1]$ denote the treatment intensity. A value close to zero means that Ukraine already had limited effective monetary sovereignty, so Euro Area membership would not have changed the monetary regime substantially. A value close to one means that Ukraine was actively using monetary sovereignty, especially through exchange-rate adjustment, so Euro Area membership would have removed a major adjustment channel.

Table 3: Treatment intensity implied by Part A monetary-regime chronology

Period	De facto regime	Treatment intensity
2000–Jul. 2008	Conventional peg / stabilised arrangement	0.15
Aug. 2008–Feb. 2009	Forced devaluation / managed float	0.85
Mar. 2009–Oct. 2013	Re-peg around 8 UAH/USD	0.15
Nov. 2013–Feb. 2014	Peg under stress	0.50
Mar. 2014–Dec. 2015	Managed float / hryvnia collapse	1.00
2016–2021	Inflation targeting and managed float	0.80
Jan. 2022–Jul. 2022	Wartime fixed exchange rate	0.20
Aug. 2022–Oct. 2023	Wartime re-peg after official devaluation	0.30
Nov. 2023–Dec. 2025	Managed flexibility with interventions	0.60

This calibration follows the fear-of-floating interpretation of [Calvo and Reinhart \(2002\)](#): the effective cost of joining the Euro Area depends on how much monetary autonomy Ukraine was actually exercising in each period.

2.4 Baseline method: ARDL / Local Projection

The baseline specification models Ukraine’s year-on-year inflation as a function of inflation persistence, the Euro Area common inflation factor, and exchange-rate depreciation:

$$\pi_t^{UA} = \alpha + \rho_1 \pi_{t-1}^{UA} + \rho_2 \pi_{t-2}^{UA} + \gamma_0 F_t^{EA} + \gamma_1 F_{t-1}^{EA} + \delta_0 \Delta e_t^{yoy} + \delta_1 \Delta e_{t-1}^{yoy} + u_t, \quad (3)$$

where π_t^{UA} is Ukraine’s actual year-on-year inflation, F_t^{EA} is the PCA-based Euro Area common inflation factor, and Δe_t^{yoy} is the year-on-year depreciation of the hryvnia against the euro.

The identifying assumption is that Euro Area membership would have affected Ukraine’s inflation through two main channels. First, Ukraine would have imported the Euro Area nominal anchor, approximated by the common Euro Area inflation factor. Second, Ukraine would not have had an independent hryvnia/euro exchange rate. Therefore, the domestic depreciation channel is removed proportionally to the treatment intensity:

$$\Delta e_t^{cf} = (1 - w_t) \Delta e_t^{yoy}. \quad (4)$$

The counterfactual inflation path is generated dynamically. Rather than using actual lagged inflation in every period, the recursion uses lagged simulated counterfactual inflation:

$$\pi_t^{cf} = \hat{\alpha} + \hat{\rho}_1 \pi_{t-1}^{cf} + \hat{\rho}_2 \pi_{t-2}^{cf} + \hat{\gamma}_0 F_t^{EA} + \hat{\gamma}_1 F_{t-1}^{EA} + \hat{\delta}_0 \Delta e_t^{cf} + \hat{\delta}_1 \Delta e_{t-1}^{cf}. \quad (5)$$

This dynamic recursion is important because it allows the counterfactual path to evolve endogenously over time rather than applying only a one-period correction to actual inflation.

2.5 Baseline ARDL/LP results

The ARDL/LP model is estimated at monthly frequency from February 2001 to December 2025. The model fits Ukraine’s inflation dynamics well, with an R^2 of approximately 0.98. Inflation persistence is strong and statistically significant. The exchange-rate depreciation channel is also economically important, confirming that hryvnia depreciation contributed to Ukrainian inflation surges during crisis episodes.

Table 4: Baseline ARDL/LP counterfactual results

Period	Actual mean (%)	Counterfactual mean (%)	Gap (pp)	N
Full sample	11.8	8.4	3.4	299
GFC 2008–09	18.3	14.3	4.0	17
Donbas 2014–15	30.3	8.3	22.0	24
War 2022–23	17.0	17.2	-0.2	23
Managed float 2024–25	9.6	4.7	5.0	24

The results suggest that Euro Area membership would have reduced Ukrainian inflation mainly during episodes of large hryvnia depreciation. The effect is moderate during the Global Financial Crisis, but very large during the 2014–2015 Crimea/Donbas crisis. By contrast, the estimated gap during 2022–2023 is small because the wartime fixed exchange-rate regime already constrained Ukraine’s monetary sovereignty. In that sense, the 2022 regime was closer to an emergency fixed-exchange-rate arrangement than to a fully independent float.

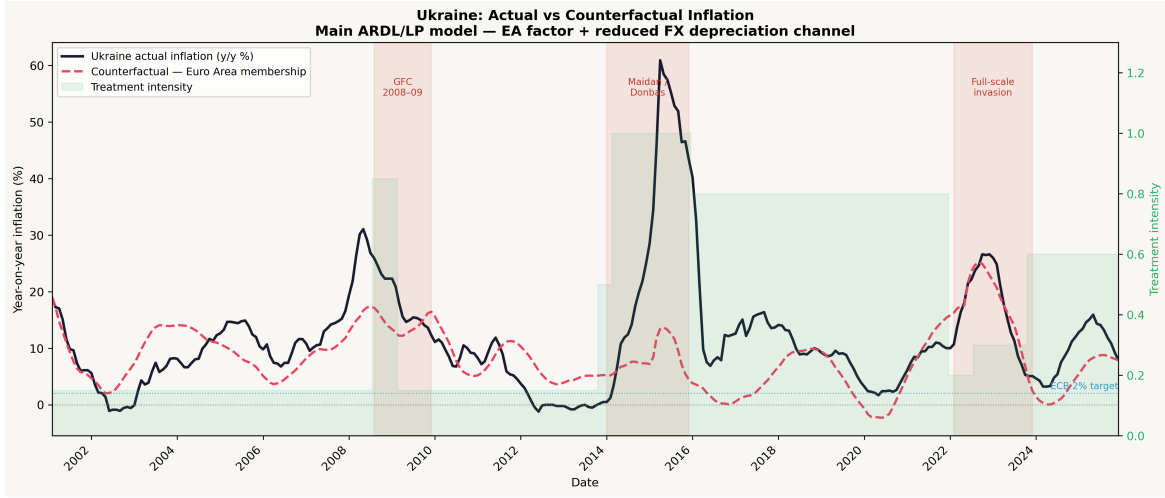


Figure 4: Ukraine: actual inflation and ARDL/LP counterfactual inflation

2.6 Structural robustness: BQ-SVAR

As a structural robustness check, I estimate a bivariate Blanchard–Quah SVAR (Blanchard and Quah, 1989), in the spirit of the optimum-currency-area shock-comparison exercise of Bayoumi and Eichengreen (1993). The SVAR is estimated separately for Ukraine and the Euro Area using:

$$Y_t = \begin{bmatrix} \Delta y_t \\ \pi_t \end{bmatrix}, \quad (6)$$

where Δy_t is industrial-production year-on-year growth and π_t is year-on-year inflation.

The identifying restriction is that demand shocks have no long-run effect on activity, while supply shocks may have permanent effects on activity. Formally, if $C(1)$ is the long-run impact matrix of structural shocks, the restriction imposes:

$$C_{12}(1) = 0. \quad (7)$$

The economic interpretation is straightforward. Supply shocks capture persistent real disruptions, such as energy shocks, production losses, war damage, or structural output shocks. These shocks are preserved as Ukraine-specific shocks in the counterfactual. Demand shocks capture nominal and demand-side disturbances. Under the Euro Area counterfactual, Ukraine’s domestic demand shocks are replaced by Euro Area demand shocks, weighted by the treatment intensity from Part A:

$$\varepsilon_t^{cf,d} = (1 - w_t)\varepsilon_t^{UA,d} + w_t\varepsilon_t^{EA,d}. \quad (8)$$

Ukraine’s supply shocks are kept unchanged:

$$\varepsilon_t^{cf,s} = \varepsilon_t^{UA,s}. \quad (9)$$

This robustness exercise implements the structural logic directly: Euro Area membership does not eliminate Ukraine-specific supply shocks, but it replaces part of the domestic nominal-demand component with Euro Area-wide demand conditions.

2.7 BQ-SVAR results

The BQ-SVAR is estimated on the common activity-inflation sample from January 2007 to September 2025.

Table 5: BQ-SVAR robustness results

Period	Actual mean inflation (%)	Counterfactual mean inflation (%)	Mean gap (pp)	N
Full sample	13.0	10.8	2.2	225
GFC 2008–09	18.3	10.5	7.8	17
Donbas 2014–15	30.3	6.6	23.7	24
War 2022–23	17.0	26.2	-9.2	23
Managed float 2024–25	9.7	7.9	1.8	21

The BQ-SVAR confirms the baseline result for the two major devaluation episodes. In 2008–2009 and especially in 2014–2015, counterfactual inflation is substantially below

actual inflation. This supports the interpretation that monetary sovereignty was costly in nominal terms when it operated through sharp exchange-rate depreciation.

The 2022–2023 BQ-SVAR result differs: counterfactual inflation is higher than actual inflation. This reflects the fact that Euro Area demand and inflation shocks were themselves strong during the European energy crisis following Russia’s invasion of Ukraine. Thus, replacing Ukrainian demand shocks with Euro Area demand shocks during this period can generate a more inflationary counterfactual. This does not invalidate the approach; it highlights that Euro Area membership would not have insulated Ukraine from common European energy-price shocks.

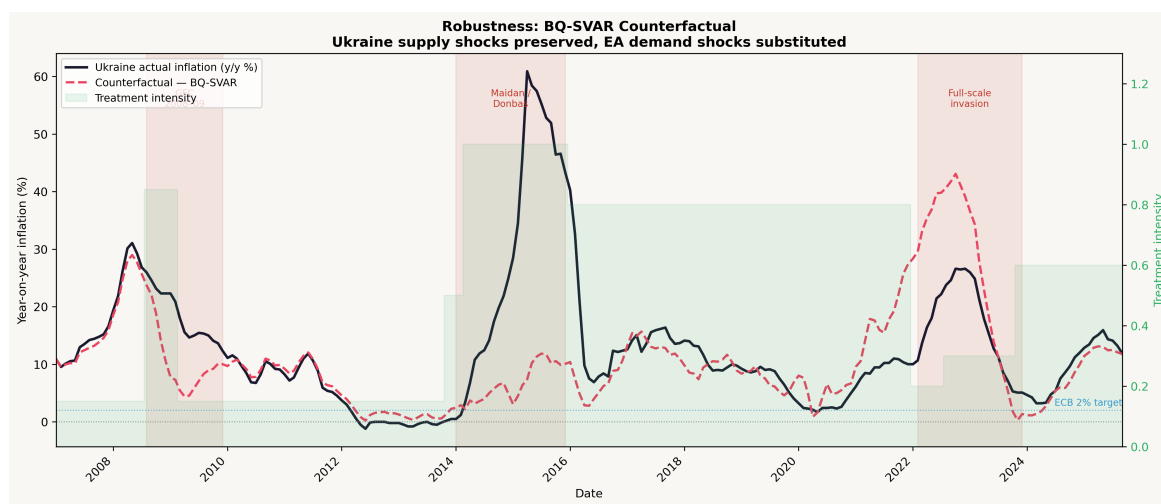


Figure 5: Ukraine: actual inflation and BQ-SVAR counterfactual inflation

2.8 Additional robustness: synthetic control

As an additional robustness exercise, I compute a synthetic-control counterfactual using an extended panel of European inflation series. The synthetic-control method constructs a weighted combination of donor countries that best approximates Ukraine’s pre-crisis inflation dynamics (Abadie et al., 2010). This method is not used as the main result because it is less directly linked to the SVAR/local-projection toolkit required by the exam, but it provides a useful non-parametric benchmark.

Table 6: Synthetic-control robustness results

Period	Actual mean inflation (%)	Counterfactual mean inflation (%)	Mean gap (pp)	N
Full sample	11.8	4.9	6.9	298
GFC 2008–09	18.3	6.2	12.1	17
Donbas 2014–15	30.3	0.5	29.8	24
War 2022–23	17.0	13.1	3.9	23
Managed float 2024–25	9.7	3.3	6.4	23

The synthetic-control results are directionally consistent with the baseline ARDL/LP and BQ-SVAR findings for 2008–2009 and 2014–2015, although the magnitude is larger. This is expected because synthetic control does not explicitly preserve Ukraine-specific supply shocks and may therefore overstate the disinflationary effect of Euro Area membership during severe geopolitical crises.

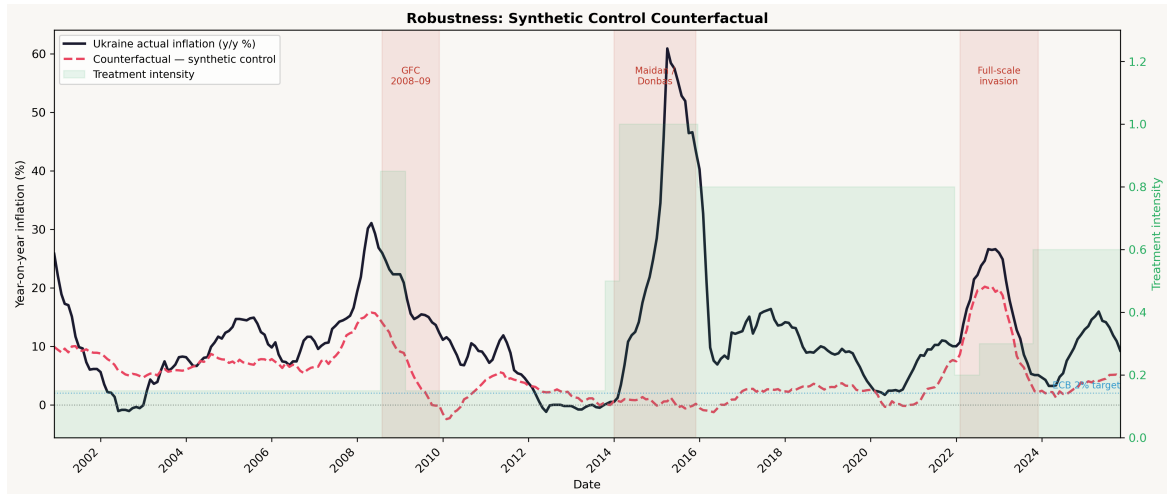


Figure 6: Ukraine: actual inflation and synthetic-control counterfactual inflation

2.9 Comparison across methods

Table 7 compares the three counterfactual methods. The central result is robust: all methods imply that Euro Area membership would have reduced inflation during Ukraine’s major devaluation episodes, especially in 2014–2015.

Table 7: Comparison of counterfactual results across methods

Method	Period	Actual mean inflation (%)	Counterfactual mean inflation (%)	Mean gap (pp)	N
Main ARDL/LP	Full sample	11.8	8.4	3.4	299
Main ARDL/LP	GFC 2008–09	18.3	14.3	4.0	17
Main ARDL/LP	Donbas 2014–15	30.3	8.3	22.0	24
Main ARDL/LP	War 2022–23	17.0	17.2	-0.2	23
Main ARDL/LP	Managed float 2024–25	9.6	4.7	5.0	24
BQ-SVAR robustness	Full sample	13.0	10.8	2.2	225
BQ-SVAR robustness	GFC 2008–09	18.3	10.5	7.8	17
BQ-SVAR robustness	Donbas 2014–15	30.3	6.6	23.7	24
BQ-SVAR robustness	War 2022–23	17.0	26.2	-9.2	23
BQ-SVAR robustness	Managed float 2024–25	9.7	7.9	1.8	21
Synthetic control robustness	Full sample	11.8	4.9	6.9	298
Synthetic control robustness	GFC 2008–09	18.3	6.2	12.1	17
Synthetic control robustness	Donbas 2014–15	30.3	0.5	29.8	24
Synthetic control robustness	War 2022–23	17.0	13.1	3.9	23
Synthetic control robustness	Managed float 2024–25	9.7	3.3	6.4	23

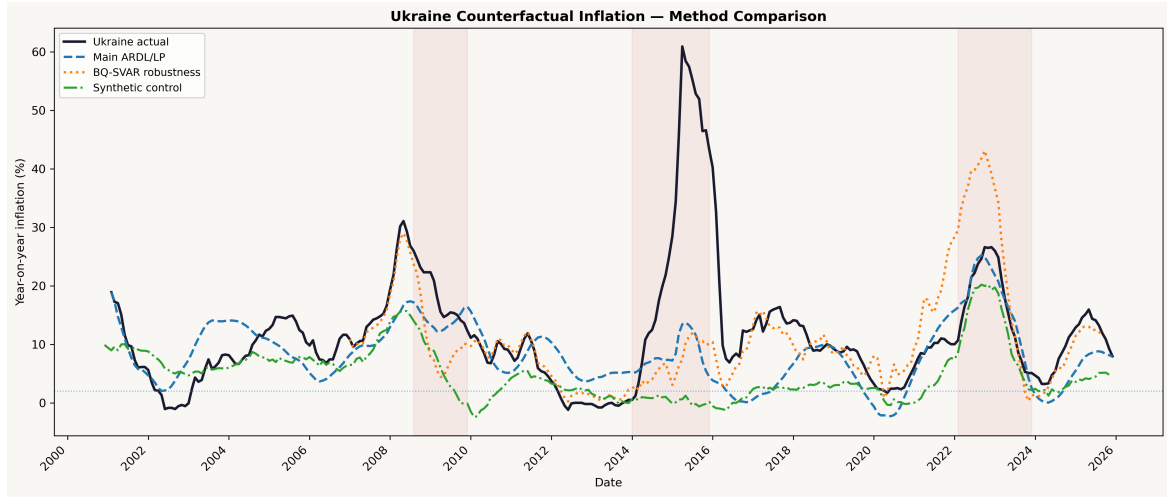


Figure 7: Comparison of counterfactual inflation paths across methods

2.10 Interpretation

The counterfactual suggests that Euro Area membership would have reduced Ukraine's inflation volatility mainly by weakening the domestic exchange-rate depreciation channel. The effect is visible during the 2008–2009 Global Financial Crisis and becomes very large during the 2014–2015 Crimea/Donbas crisis, when the collapse of the hryvnia generated strong exchange-rate pass-through into consumer prices. During 2022–2023, the baseline ARDL/LP gap is much smaller because the wartime fixed exchange-rate regime already constrained Ukraine's monetary sovereignty. The BQ-SVAR robustness check shows that Euro Area shocks were themselves inflationary during the European energy crisis, which

explains why the structural counterfactual can exceed actual Ukrainian inflation in 2022–2023. Overall, the results imply that monetary sovereignty was costly in nominal terms during devaluation episodes, but it also provided an exchange-rate adjustment mechanism that Euro Area membership would have removed.

2.11 Limitations

The counterfactual should not be interpreted as showing that Euro Area membership would have been costless. The models mainly capture inflation dynamics and do not fully model the real-side adjustment costs that would have occurred without an independent exchange rate. In particular, if Ukraine had been unable to devalue during 2008–2009 or 2014–2015, adjustment would likely have occurred through output losses, unemployment, fiscal contraction, wage compression, or sovereign-risk channels. This is the classic optimum-currency-area trade-off: Euro Area membership can reduce inflation volatility and import credibility, but it also removes the exchange rate as a shock absorber (Mundell, 1961; McKinnon, 1963; Kenen, 1969; De Grauwe, 2012).

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